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Centre Number

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Student Number

SCEGGS Darlinghurst

2007

Higher School Certificate
Assessment Task 2

Mathematics-Extension I

Task Weighting: 35%

Outcomes Assessed: HE3, HE4, HE6 & HE7

General Instructions

- Time allowed – 60 minutes
- Start each question on a new page.
- Attempt all questions and show all necessary working.
- Answer **Question 1 (c)** on the answer sheet provided
- Write your student number at the top of each page.
- Marks can be deducted for careless or badly arranged work
- Mathematical templates, geometrical equipment and scientific calculators may be used.

Question	Reasoning	Communication	Total
1	/2	/4	/12
2	/7	/1	/13
3	/3	/3	/13
4	/7	/1	/12
Total	/19	/9	/50

Calc

/6

/3

/2

/11

Average: _____

St. Dev.: _____

Rank: _____

Parent's Signature _____

Start A New Page

Marks

Question 1: (12 marks)

(a) Find:

(i) $\int \frac{dx}{\sqrt{16-x^2}}$

2

(ii) $\int x(x-2)^5 dx$ using the substitution $u = x-2$

2

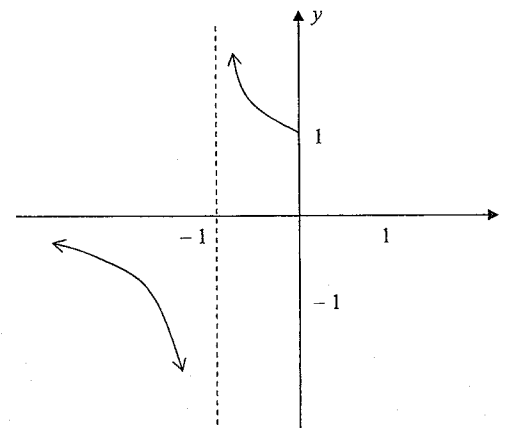
(b) Evaluate:

$$\int_{-1/4}^{1/2} \frac{dx}{\sqrt{1-4x^2}}$$

2

(c) The graph of $y = f(x)$ is drawn below:

2



On the answer sheet provided sketch $y = f^{-1}(x)$

Question 1 continued on the next page

Marks

Question 1 (continued)

- (d) The half life of a radioactive substance is the time it takes a given amount of the substance to lose one half of its mass. It is given that the half-life of plutonium-239 is 24 000 years.

Assume that plutonium-239 decays according to the law:

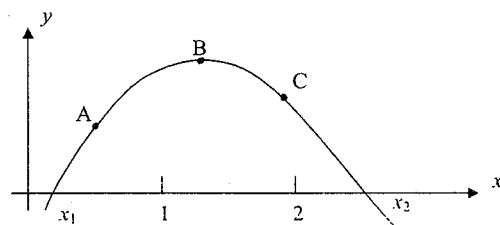
$$M = M_0 e^{-kt} \quad \text{where } M = \text{mass}$$

t = time in years

M_0 and k are constants

Find how long it would take for an amount of plutonium-239 to lose 80% of its mass. (Answer to nearest year)

- (e) Consider the graph of $y = f(x)$



Siobhan wants to calculate the value of x_1 . It is the root of the equation $f(x) = 0$ between 0 and 1. She uses Newton's Method.

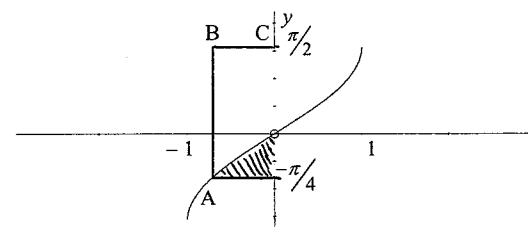
- (i) Explain, graphically, why the x - value of A is a better choice than the x - value of C as a first approximation of x_1 . 1
- (ii) Explain what would happen if Siobhan used the x - value of B. 1

Marks

Question 2: (13 marks)

- (a) $x^4 - 10x + 7 = 0$ has a root between 0.6 and 0.9. Use halving the interval method twice to show the root lies between 0.675 and 0.75. 2

- (b) The diagram shows a sketch of the function $y = \sin^{-1} x$



- (i) What are the coordinates of B? 1
- (ii) Show the area of the shaded region is $\frac{2-\sqrt{2}}{2}$ units². 3
- (iii) Hence calculate the area bounded by $y = \sin^{-1} x$, the y - axis and the intervals AB and BC 1
- (c) An ice cube tray is filled with water at a temperature of 18°C and placed in a freezer that has a constant temperature of -19°C . The cooling rate of the water is proportional to the difference between the temperature of the freezer and the temperature of the water T .
- T satisfies the equation $\frac{dT}{dt} = k(T + 19)$
- (i) Show that $T = -19 + Ae^{-kt}$ satisfies the equation for $\frac{dT}{dt}$ and find the value of A. 2
- (ii) After 5 minutes in the freezer the temperature of the water 3°C . Find the time for the water to reach -18.9°C . 3
- (iii) Sketch a graph of Temperature versus Time labelling all important features. 1

Start A New Page

Marks

Question 3: (13 marks)

- (a) (i) Show that the equation $e^x = x + 2$ has a solution in the interval $1 < x < 2$. 1

- (ii) Letting $x_1 = 1.5$ use one application of Newton's Method to approximate the solution to 3 decimal places. 3

- (b) By using the substitution $x = \tan \theta$ evaluate $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{dx}{(1+x^2)^{3/2}}$ 3

- (c) Consider the function $y = \cos^{-1}(x-1)$.

- (i) Find the domain of the function. 1

- (ii) Sketch the graph of the curve $y = f(x)$ showing clearly the coordinates of the endpoints. 2

- (iii) The region in the first quadrant bounded by the curve $y = f(x)$ and the coordinate axes is rotated about the y -axis. Find the exact value of the volume of the solid of revolution. 3

Start A New Page

Marks

Question 4: (12 marks)

- (a) Find the exact value of $\sin \left[\tan^{-1} \left(\frac{-1}{2} \right) + \cos^{-1} \left(\frac{2}{3} \right) \right]$ 3

- (b) (i) Show that $\frac{u}{u+1} = 1 - \frac{1}{u+1}$ 1

- (ii) Hence find $\int \frac{dx}{1+\sqrt{x}}$ using the substitution $x = u^2$ 2

- (c) $y = \sinh x$ is an example of a hyperbolic function. It is defined as:

$$\sinh x = \frac{1}{2}(e^x - e^{-x})$$

- (i) Find $\frac{dy}{dx}$ 1

- (ii) Explain why $y = \sinh x$ has an inverse function 1

- (iii) Show that $f^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$ for all x . 4

END OF PAPER

Start A New Page

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Mathematics Extension 1 - Assessment Task 2 HSC 2007 - Solutions

2. a) i) $\int \frac{dx}{\sqrt{16 - x^2}} = \sin^{-1} \frac{x}{4} + C$ Calc-2

2. ii) $\int x(x-2)^5 dx$ let $u = x-2$
 $du = 1 \cdot dx$
 $= \int (u+2) u^5 du$ ✓
 $= \int u^6 + 2u^5 du$
 $= \frac{u^7}{7} + \frac{2u^6}{6} + C$
 $= \frac{(x-2)^7}{7} + \frac{(x-2)^6}{3} + C$ ✓ Calc-2

Remember to change back to x !

2. b) $\int_{-\frac{1}{4}}^{\frac{1}{2}} \frac{dx}{\sqrt{1-4x^2}}$
 $= \frac{1}{2} \int_{-\frac{1}{4}}^{\frac{1}{2}} \frac{dx}{\sqrt{\frac{1}{4} - x^2}}$
 $= \frac{1}{2} [\sin^{-1} 2x]_{-\frac{1}{4}}^{\frac{1}{2}}$ ✓
 $= \frac{1}{2} (\sin^{-1} 1 - \sin^{-1} (-\frac{1}{2}))$
 $= \frac{1}{2} (\frac{\pi}{2} - (-\frac{\pi}{6}))$
 $= \frac{\pi}{3}$ ✓ Calc-2

2. c) Refer to answer sheet ✓✓

Communication - 2

Intersection must be on the line $y = x$.

2 d) find k if $M = \frac{M_0}{2}$ when $t = 24000$

$$\begin{aligned} \therefore \frac{M_0}{2} &= M_0 e^{kt} \\ \frac{1}{2} &= e^{kt} \\ \ln \frac{1}{2} &= 24000 \times k \\ k &= \frac{\ln(\frac{1}{2})}{24000} \end{aligned}$$

$$= -2.89 \times 10^{-5} \text{ (3 sig fig)}$$

find t when $M = 0.2M_0$

$$\begin{aligned} \therefore 0.2M_0 &= M_0 e^{kt} \\ 0.2 &= e^{kt} \\ \ln(0.2) &= kt \\ t &= \frac{\ln(0.2)}{k} \\ &= \frac{\ln(0.2)}{\ln(\frac{1}{2})} \times 24000 \end{aligned}$$

$$= 55726 \text{ (to nearest whole no.)}$$

$\therefore$ It will take 55726 years to lose 80% of mass

Reasoning - 2

note: $M \neq 0.8M_0$!!

1 e) i) the tangent at A cuts the x-axis closer to x_1 than the tangent at C.

Communication - 1

1 a) the gradient of the tangent is zero
 $\therefore f'(x) = 0 \therefore \frac{f(x)}{f'(x)}$ is undefined and

Newton's Method does not work.

or

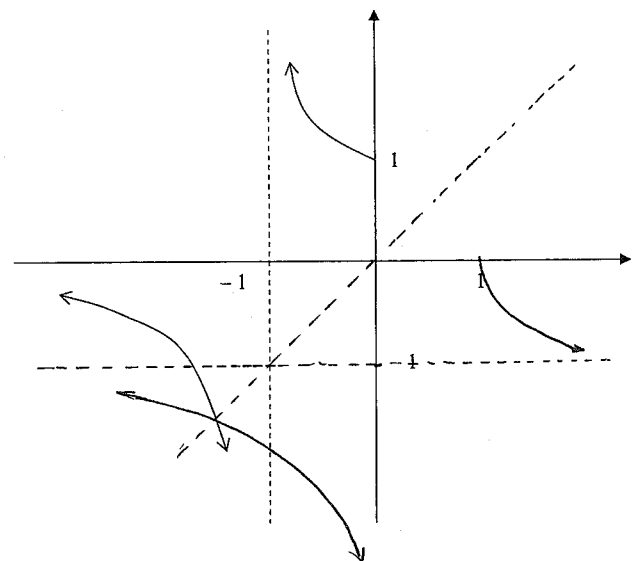
The tangent does not cut the x-axis.

Communication - 1

draw a diagram
must mention the
tangent and that
the new approximation
is where the tangent
crosses the x-axis.

Start A New Page

Answer Sheet for Question 1 (c)



Q2 a) $f(x) = x^4 - 10x + 7$

✓ $f(0.6) = 1.1296$

$f(0.9) = -1.3439$

$\frac{0.6+0.9}{2} = 0.75$

$f(0.75) = -0.18359...$ ✓

∴ root lies between 0.6 and 0.75

$\frac{0.6+0.75}{2} = 0.675$

$f(0.675) = 0.45759...$ ✓

∴ root lies between 0.675 and 0.75

✓ 1 b) i) $B(-\frac{1}{\sqrt{2}}, \frac{\pi}{2})$ ✓

✓ 3 ii) $y = \sin^{-1}x$

$x = \sin y$

Area = $\left| \int_{-\frac{\pi}{4}}^0 \sin y dy \right|$ ✓

$= \left| [-\cos y]_{-\frac{\pi}{4}}^0 \right|$

$= \left| (-\cos 0) - (-\cos(-\frac{\pi}{4})) \right|$ ✓

$= \left| -1 - (-\frac{1}{\sqrt{2}}) \right|$

$= \left| \frac{1}{\sqrt{2}} - 1 \right|$

$= \left| \frac{1-\sqrt{2}}{\sqrt{2}} \right|$

$= \left| \frac{\sqrt{2}-2}{2} \right|$ ✓

$= \frac{2-\sqrt{2}}{2} \text{ units}^2$

Reasoning - 3

✓ 1 iii) Area = $\frac{1}{\sqrt{2}} \times \frac{3\pi}{4} - \left(\frac{2-\sqrt{2}}{2} \right)$ ✓ Reasoning - 1

$= \frac{3\sqrt{2}\pi}{8} - \left(\frac{2-\sqrt{2}}{2} \right)$

To draw conclusions
you must state the
value of $f(0.6)$

✓ $f(0.9)$

because it is a 'slow'
question you must
state the value of
 $f(0.75)$ ✓ $f(0.675)$

done well

some interesting statements
here!

because it is a slow
question you must be
particular about how
you find this area

eg: why ~~do~~ these
work:

• $\int_0^{\frac{\pi}{4}} \sin y dy$

• $-\int_0^{\frac{\pi}{4}} \sin y dy$

etc

$= \frac{1}{8} (3\sqrt{2}\pi + 4\sqrt{2} - 6) \text{ units}^2$

✓ 2 c) i) $T = -19 + Ae^{kt}$

$\frac{dT}{dt} = kAe^{kt}$

∴ LHS = $\frac{dT}{dt}$
 $= kAe^{kt}$

RHS = $k(T+19)$
 $= k(-19 + Ae^{kt} + 19)$ ✓
 $= kAe^{kt}$

∴ LHS = RHS

∴ $T = -19 + Ae^{kt}$ satisfies the equation

when $t=0$ $T=18$

$18 = -19 + Ae^{k \times 0}$

$37 = Ae^0$

$A = 37$ ✓

done well

✓ 3 ii) find k : $T=3$ $t=5$

∴ $3 = -19 + 37e^{k \times 5}$

$22 = 37e^{k \times 5}$

$\frac{22}{37} = e^{k \times 5}$

$\ln\left(\frac{22}{37}\right) = k \times 5$

$k = \frac{1}{5} \ln\left(\frac{22}{37}\right)$ ✓ $(-0.1039...)$

∴ find t when $T = -18.9$

$-18.9 = -19 + 37e^{k \times t}$ ✓

$0.1 = 37e^{k \times t}$

$\frac{0.1}{37} = e^{k \times t}$

$\ln\left(\frac{0.1}{37}\right) = k \times t$

$t = \frac{\ln\left(\frac{0.1}{37}\right)}{k}$

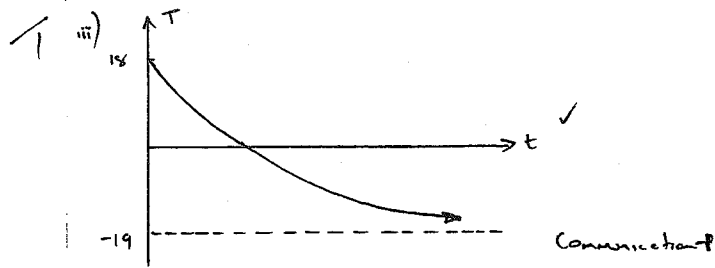
$= 56.87...$

∴ 57 minutes ✓

Reasoning - 3

done well by nearly
all candidates.

Errors were made
by using incorrect
values for T & t



Q3 a) i) $e^x = x + 2 \rightarrow e^x - x - 2 = 0$

let $f(x) = e^x - x - 2$

$f(1) = e^1 - 1 - 2 = -0.281... < 0$ $f(2) = e^2 - 2 - 2 = 3.389... > 0$ ✓

Since $f(1) < 0$ and $f(2) > 0$ and $f(x)$ is continuous then $f(x)$ has a root between $x=1$ and $x=2$.
Communication - 1

3 ii) $f'(x) = e^x - 1$

$x_2 = 1.5 - \frac{f(1.5)}{f'(1.5)}$ ✓

$= 1.5 - \frac{e^{1.5} - 1.5 - 2}{e^{1.5} - 1}$ ✓

$= 1.218$ (to 3 dec. pl.) ✓
(correct rounding)

3 b) $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{dx}{(1+x^2)^{3/2}}$

$x = \tan \theta$ $x = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$

$dx = \sec^2 \theta d\theta$ $x = \frac{1}{\sqrt{3}} \Rightarrow \theta = \frac{\pi}{6}$

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 \theta d\theta}{(1+\tan^2 \theta)^{3/2}}$ ✓

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 \theta d\theta}{(\sec^2 \theta)^{3/2}}$

curve must be continuous

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{d\theta}{\sec \theta}$

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos \theta d\theta$ ✓

$= \sin \theta \Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}}$

$= \sin \frac{\pi}{3} - \sin \frac{\pi}{6}$

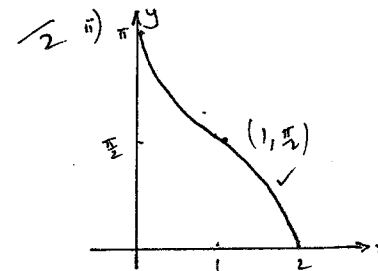
$= \frac{\sqrt{3}}{2} - \frac{1}{2}$

$= \frac{\sqrt{3}-1}{2}$ ✓

Calc - 3

1 c) i) $-1 \leq x-1 \leq 1$

$0 \leq x \leq 2$ ✓



1- for graph
1- for labelling

Communication - 2

3 iii) $y = \cos^{-1}(x-1)$

$\cos y = x-1$

$x = \cos y + 1$

$V = \pi \int_0^{\pi} (\cos y + 1)^2 dy$ ✓

$= \pi \int_0^{\pi} \cos^2 y + 2\cos y + 1 dy$

$= \pi \int_0^{\pi} \frac{1}{2} \cos 2y + 2\cos y + \frac{3}{2} dy$

now $\cos 2y = 2\cos^2 y - 1$
 $\cos^2 y = \frac{1}{2}(1 + \cos 2y)$

$$= \pi \left[\frac{1}{4} \sin 2y + 2 \sin y + \frac{3}{2} y \right]_0^{\pi} \quad \checkmark$$

$$= \pi \left[\left(\frac{1}{4} \sin 2\pi + 2 \sin \pi + \frac{3}{2} \pi \right) - (0 + 0 + 0) \right]$$

$$= \pi \times \frac{3\pi}{2}$$

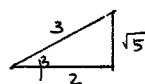
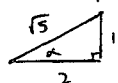
$$= \frac{3\pi^2}{2} \text{ units}^2 \quad \checkmark$$

Reasoning - 3

Q4 a) let $\alpha = \tan^{-1}\left(-\frac{1}{2}\right)$ $\beta = \cos^{-1}\left(\frac{2}{3}\right)$

$\tan \alpha = -\frac{1}{2}$ $\checkmark$ $\cos \beta = \frac{2}{3}$

$\therefore \alpha$ is in 4th quad.



$$\therefore \sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \quad \checkmark$$

$$= \frac{-1}{\sqrt{5}} \times \frac{2}{3} + \frac{2}{\sqrt{5}} \times \frac{\sqrt{5}}{3}$$

$$= \frac{-2}{3\sqrt{5}} + \frac{2}{3}$$

$$= \frac{-2\sqrt{5}}{15} + \frac{2}{3}$$

$$= \frac{10 - 2\sqrt{5}}{15} \quad \checkmark$$

Reasoning - 3

1 b) i) $RHS = 1 - \frac{1}{u+1}$

$$= \frac{u+1-1}{u+1} \quad \checkmark$$

$$= \frac{u}{u+1}$$

2 ii) $\int \frac{du}{1+\sqrt{u}}$ $u = u^2$ $du = 2u du$

$$= \int \frac{2u du}{1+u} \quad \checkmark$$

$$= 2 \int \frac{u du}{1+u}$$

$$= 2 \int \left(1 - \frac{1}{u+1} \right) du$$

$$= 2 \left(u - \ln|u+1| \right) + c$$

Calc - 2

You must remember to replace u with $\sqrt{x}$ after integrating.

$$= 2\sqrt{x} - 2\ln(1+\sqrt{x}) + c \quad \checkmark$$

1 c) i) $\frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{2} (e^x - e^{-x}) \right)$

$$= \frac{1}{2} (e^x + e^{-x}) \quad \checkmark$$

1 ii) since $e^x > 0$ and $e^{-x} > 0$ for all x

$$\frac{dy}{dx} > 0 \text{ for all } x$$

$\therefore y = \sinh x$ is a monotonic increasing function

$\therefore y = \sinh x$ has an inverse fn. $\checkmark$ (communication)

4 iii) $y = \frac{1}{2} (e^x - e^{-x})$

interchange x and y

$$x = \frac{1}{2} (e^y - e^{-y}) \quad \checkmark$$

$$2x = e^y - e^{-y}$$

$$2xe^y = e^{2y} - 1$$

$$e^{2y} - 2xe^y - 1 = 0 \quad \checkmark$$

$$\text{let } m = e^y$$

$$m^2 - 2xm - 1 = 0$$

$$m = \frac{2x \pm \sqrt{(2x)^2 - 4 \cdot 1 \cdot (-1)}}{2}$$

$$= \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$= \frac{2x \pm 2\sqrt{x^2 + 1}}{2}$$

$$= x \pm \sqrt{x^2 + 1} \quad \checkmark$$

$$\therefore e^y = x \pm \sqrt{x^2 + 1}$$

$$\text{since } \sqrt{x^2 + 1} > x \quad \checkmark$$

$$e^y = x + \sqrt{x^2 + 1}$$

$$y = \ln(x + \sqrt{x^2 + 1})$$

Reasoning - 4

Done well

Very poor reasoning here. You must link $\frac{dy}{dx} > 0$ with monotonic increasing

Very few candidates know how to progress past $x = \frac{1}{2} (e^y - e^{-y})$